Unit outline_

COMP5310: Principles of Data Science

Semester 2, 2026 [Normal evening] - Camperdown/Darlington, Sydney

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Overview –

The focus of this unit is on understanding and applying relevant concepts, techniques, algorithms, and tools for the analysis, management and visualisation of data- with the goal of enabling discovery of information and knowledge to guide effective decision making and to gain new insights from large data sets. To this end, this unit of study provides a broad introduction to data management, analysis, modelling and visualisation using the Python programming language. Development of custom software using the powerful, general-purpose Python scripting language; Data collection, cleaning, pre-processing, and storage using various databases; Exploratory data analysis to understand and profile complex data sets; Mining unlabelled data to identify relationships, patterns, and trends; Machine learning from labelled data to predict into the future; Communicate findings to varied audiences, including effective data visualisations. Core data science content will be taught in normal lecture + tutorial delivery mode. Python programming will be taught through an online learning platform in addition to the weekly face-to-face lecture/tutorials. The unit of study will include hands-on exercises covering the range of data science skills above.

Unit details and rules

Academic unit	Computer Science
Credit points	6
Prerequisites ?	None
Corequisites ?	None
Prohibitions ?	INFO3406 or OCMP5310
Assumed knowledge ?	Good understanding of relational data model and database technologies as covered in ISYS2120 or COMP9120 (or equivalent UoS from different institutions)
Available to study abroad and exchange students	Yes

Teaching staff

Coordinator	Nguyen Tran, nguyen.tran@sydney.edu.au

Assessment –

ⓘ The census date for this unit availability is **31 August 2026**

Details | Criteria

Type	Description	Weight	Due	Length	Use of AI
Written exam 🚧	**Written exam** Supervised Final exam	60%	Formal exam period	2 hours	AI prohibited
Outcomes assessed: LO1 LO2 LO3 LO4 LO5 LO8 LO9 LO10					
Out-of-class quiz ❗	**Homework Quiz** In-semester quiz.	5%	Week 03 **Due date**: 18 Aug 2026 at 17:00 **Closing date**: 21 Aug 2026	n/a	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4					
Data analysis 👥	**Assignment 1: Obtain data, clean it, load, summarise and analyse the data.** Computer program and report.	15%	Week 06 **Due date**: 11 Sep 2026 at 23:59 **Closing date**: 18 Sep 2026	n/a	AI allowed
Outcomes assessed: LO2 LO6 LO7 LO8					
Data analysis 👥	**Assignment 2: Develop and evaluate predictive model.** Computer program and report.	20%	Week 11 **Due date**: 23 Oct 2026 at 23:59 **Closing date**: 30 Oct 2026	n/a	AI allowed
Outcomes assessed: LO1 LO6 LO7 LO8					

🚧 = hurdle task ? 👥 = group assignment ? ❗ = early feedback task ?

Assessment summary

- **Homework quiz:** out-of-class quiz.
- **Assignment 1:** Obtain data, clean it, load, summarise and analyse the data.
- **Assignment 2:** Develop and evaluate predictive model.
- **Exam** – supervised exam during exam period
- **Exam requirement**: a minimum of 40% at the exam is required to pass this course

Detailed information for each assessment can be found on Canvas. Ethical AI usage, critical thinking, and originality are expected in all assignments.

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, for assessments in this unit and check Canvas for full instructions on assessment tasks and AI use.

If you use AI, you must always acknowledge it. Misusing AI may lead to a breach of the Academic Integrity Policy.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Late submission

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

This unit has an exception to the standard University policy or supplementary information has been provided by the unit coordinator. This information is displayed below:

Homework quiz, Assignment 1, and Assignment 2: Late submissions are allowed up to 3 days late. A penalty of 5% per day late will apply.

Academic integrity

The University expects students to act ethically and honestly and will treat all allegations of academic integrity breaches seriously.

Our website provides information on academic integrity and the resources available to all students. This includes advice on how to avoid common breaches of academic integrity. Ensure that you have completed the Academic Honesty Education Module (AHEM) which is mandatory for all commencing coursework students

Penalties for serious breaches can significantly impact your studies and your career after graduation. It is important that you speak with your unit coordinator if you need help with completing assessments.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Learning support –

Simple extensions

If you encounter a problem submitting your work on time, you may be able to apply for an extension of five calendar days through a simple extension. The application process will be different depending on the type of assessment and extensions cannot be granted for some assessment types like exams.

Special consideration

If exceptional circumstances mean you can't complete an assessment, you need consideration for a longer period of time, or if you have essential commitments which impact your performance in an assessment, you may be eligible for special consideration or special arrangements.

Special consideration applications will not be affected by a simple extension application.

Using AI responsibly

Co-created with students, AI in Education includes lots of helpful examples of how students use generative AI tools to support their learning. It explains how generative AI works, the different tools available and how to use them responsibly and productively.

Support for students

The Support for Students Policy reflects the University's commitment to supporting students in their academic journey and making the University safe for students. It is important that you read and understand this policy so that you are familiar with the range of support services available to you and understand how to engage with them.

The University uses email as its primary source of communication with students who need support under the Support for Students Policy. Make sure you check your University email regularly and respond to any communications received from the University.

Learning resources and detailed information about weekly assessment and learning activities can be accessed via Canvas. It is essential that you visit your unit of study Canvas site to ensure you are up to date with all of your tasks.

If you are having difficulties completing your studies, or are feeling unsure about your progress, we are here to help. You can access the support services offered by the University at any time:

Support and Services (including health and wellbeing services, financial support and learning support)
Course planning and administration
Meet with an Academic Adviser

Weekly schedule –

WK	Topic	Learning activity	Learning outcomes
Week 01	Introduction to data science and big data	Lecture (2 hr)	LO3 LO4 LO8
	Introduction to data science and big data	Tutorial (1 hr)	LO3 LO4 LO8
Week 02	Data exploration with Spreadsheets	Lecture (2 hr)	LO5 LO8
	Data exploration with Spreadsheets	Tutorial (1 hr)	LO5 LO8
Week 03	Data exploration with Python	Lecture (2 hr)	LO2 LO6
	Data exploration with Python	Tutorial (1 hr)	LO2 LO6
Week 04	Cleaning and storing data	Lecture (2 hr)	LO2 LO7
	Cleaning and storing data	Tutorial (1 hr)	LO2 LO7
Week 05	Querying and summarising data	Lecture (2 hr)	LO2 LO6
	Querying and summarising data	Tutorial (1 hr)	LO2 LO6
Week 06	Hypothesis testing and evaluation	Lecture (2 hr)	LO1 LO8
	Hypothesis testing and evaluation	Tutorial (1 hr)	LO1 LO8
Week 07	Data mining: association rules and dimensionality reduction	Lecture (2 hr)	LO1 LO7 LO9 LO10
	Data mining: association rules and dimensionality reduction	Tutorial (1 hr)	LO1 LO7 LO9 LO10
Week 08	Data mining: clustering	Lecture (2 hr)	LO1 LO7 LO9 LO10
	Data mining: clustering	Tutorial (1 hr)	LO1 LO7 LO9 LO10
Week 09	Machine learning: regression	Lecture (2 hr)	LO1 LO7 LO9 LO10
	Machine learning: regression	Tutorial (1 hr)	LO1 LO7 LO9 LO10
Week 10	Machine learning: classification	Lecture (2 hr)	LO1 LO7 LO9 LO10
	Machine learning: classification	Tutorial (1 hr)	LO1 LO7 LO9 LO10
Week 11	Unstructured data	Lecture (2 hr)	LO1 LO7 LO9 LO10
	Unstructured data	Tutorial (1 hr)	LO1 LO7 LO9 LO10
Week 12	Ethics in data science	Lecture (2 hr)	
	Ethics in data science	Tutorial (1 hr)	
Week 13	Semester review	Lecture (2 hr)	LO9 LO10

Study commitment

Typically, there is a minimum expectation of 1.5-2 hours of student effort per week per credit point for units of study offered over a full semester. For a 6 credit point unit, this equates to roughly 120-150 hours of student effort in total.

Required readings

1. Géron A. Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow : Concepts, Tools, and Techniques to Build Intelligent Systems. Third edition. (ProQuest (Firm), ed.). O'Reilly; 2022.

2. Grus J. *Data Science from Scratch : First Principles with Python* . Second edition. O'Reilly; 2019.

3. Vanderplas JT. Python Data Science Handbook : Essential Tools for Working with Data . Second edition. O'Reilly Media, Incorporated; 2023.

Learning outcomes –

Learning outcomes are what students know, understand and are able to do on completion of a unit of study. They are aligned with the University's graduate qualities and are assessed as part of the curriculum.

Outcomes | Graduate qualities

At the completion of this unit, you should be able to:

LO1. select statistical techniques appropriate for evaluation of a predictive model that is based on data analysis, and justify this choice
LO2. select statistical techniques appropriate for summarisation and analysis of a data set, and justify this choice
LO3. apply concepts and terms from social science to describe and analyse the role of a data analysis task in its organisational context
LO4. understand the role of data science in decision-making
LO5. understand the technical issues that are present in the stages of a data analysis task and the properties of different technologies and tools that can be used to deal with the issues
LO6. process large data sets using appropriate technologies
LO7. carry out (in guided stages) the whole design and implementation cycle for creating a pipeline to analyse a large heterogenous dataset
LO8. seek details of how to use a method or tool in the data analytic process
LO9. communicate the results produced by an analysis pipeline, in oral and written form, including meaningful diagrams
LO10. communicate the process used to analyse a large data set, and justify the methods used.

Responding to student feedback –

This section outlines changes made to this unit following staff and student reviews.

No major changes, only updates - positive student feedback.

Disclaimer

Important: the University of Sydney regularly reviews units of study and reserves the right to change the units of study available annually. To stay up to date on available study options, including unit of study details and availability, refer to the relevant handbook.

The information included on this page is correct at 10 Jul 2026.

To help you understand common terms that we use at the University, we offer an online glossary.

We recognise and pay respect to the Elders and communities - past, present, and emerging - of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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